Toroidal Manifolds and Recursive Neural Dynamics: Grid Cell Geometry, Stability, and Information Persistence

Christopher Ezernack

Department of Physics and Cognitive Science

University of Texas at Dallas

christopher.ezernack@utdallas.edu

November 5, 2026

Abstract

Grid cell firing patterns can represent position on a torus. An attracting torus also requires nearby activity to return after a displacement. We reconstructed a population matrix from five grid cell recording files. Its 13 columns combine 23 cell recordings across trials and animals, so its rows cannot trace a simultaneous population through time. In a model of two coupled oscillators, independent phases can trace periodic or dense orbits on the torus. Coupling can lock the phases to a circle within it. Adding amplitude coordinates gives three models with identical trajectories on the torus. A small radial displacement decays in one, persists in another, and grows in the third. In the contracting model, two radial displacements encode a bit that an exact readout recovers at every finite time. A readout with fixed bin width eventually assigns both displacements to the same bin. A neural test of continuity must specify both how activity recovers and which distinctions a later observation can resolve.

1 Introduction

Grid cells fire at locations arranged approximately on a triangular lattice [1]. Relative firing rates across a population can encode position within each period of this pattern. Positions separated by a lattice period share the same coordinates. These coordinates lie on a torus. Whether displaced neural activity returns to this torus is a question about dynamics.

Simultaneous recordings provide evidence for a toroidal representation. Gardner et al. recovered toroidal population topology within grid cell modules during waking and sleep [2]. Hermansen et al. found toroidal representations during movement along a one dimensional path [3]. Yoon et al. found that relationships among cells persisted as individual responses changed [4]. Those relationships support a continuous attractor model in which nearby activity returns toward a family of represented states [5].

Spatial tuning and joint activity have different sensitivities to timing. Di Sarra et al. found that timing perturbations could disrupt the recovered topology while leaving spatial grid scores largely intact [6]. Their model of independent Poisson firing with modulated rates also produced toroidal signatures [6, 7]. A toroidal signature alone leaves the mechanism of recovery undetermined.

Two coupled oscillators provide a simple example. Neural rhythms and interactions across frequencies motivate a description in terms of phase [8–10]. Each phase specifies a position on a circle. Together, the two phases specify a point on a torus. The phase equations determine motion along that surface. Amplitude coordinates describe what happens when a displacement takes the state away from it.

The trajectories in two such models can agree everywhere on the torus even when one model attracts displaced states and the other repels them. Recovery can preserve a difference between states that approach the same limit. Whether an observer can still read that difference depends on the resolution of the measurement.

The grid cell analysis identifies which recordings share position traces and which were pooled across trials. The oscillator model gives separate predictions for phase locking, return after a displacement, and loss of a distinction at finite resolution. These predictions specify measurements relevant to the continuity of a neural representation and, in a proposed human test, to a report of conscious experience.

2 Methods

2.1 Grid Cell Recordings and Spatial Tuning

We examined five files from the Hafting grid cell recordings [1, 11]. The accompanying documentation identifies simultaneously recorded cells within each file. We kept the spike records separate for each file. To reconstruct the archived population matrix, we merged repeated cell labels on one recording's time axis, then compared every entry with the stored matrix.

For each spatial map, we assigned each bin a probability proportional to its firing rate and computed the entropy of this distribution. We also counted intervals and measured their lifetimes in the archived persistence diagrams. Appendix C specifies the recording inventory, normalization, and preprocessing.

2.2 Oscillator Model

Oscillator i has phase $\phi_i \in \mathbb{R}/2\pi\mathbb{Z}$ and constant natural frequency ω_i. The phase pair lies on $T^2 = S^1 \times S^1$. With symmetric sinusoidal coupling [12, 13], the phases satisfy

$$\begin{aligned} \dot{\phi}_1 &= \omega_1 + \epsilon \sin(\phi_2 - \phi_1), \\ \dot{\phi}_2 &= \omega_2 + \epsilon \sin(\phi_1 - \phi_2), \qquad \epsilon \geq 0. \end{aligned} \tag{1}$$

Here $\epsilon = 0$ gives independent rotations. With phase difference $\delta = \phi_2 - \phi_1$ and detuning $\Delta\omega = \omega_2 - \omega_1$, the relative dynamics reduce to

$$\dot{\delta} = \Delta\omega - 2\epsilon \sin \delta. \tag{2}$$

On a lift to real angles, the mean phase advances at $(\omega_1 + \omega_2)/2$. A stable equilibrium of Eq. (2) gives a stable phase difference.

To describe a displacement away from the torus, we add two positive amplitudes r_1, r_2 satisfying

$$\dot{r}_i = -\lambda(r_i - 1), \qquad \lambda > 0. \tag{3}$$

The resulting state is $(r_1 \cos \phi_1, r_1 \sin \phi_1, r_2 \cos \phi_2, r_2 \sin \phi_2)$ in $\mathbb{R}^4$. Both amplitudes equal one on the unit product torus. Changing an amplitude displaces the state in a direction normal to the torus. Appendix B also treats a periodic forcing of the amplitudes with an exact solution.

2.3 Simulations and Observation Rules

We simulated the phase pair at frequencies $(1, \sqrt{2})$ with coupling strengths 0, 0.1, and 0.6. A control at $(1, 3/2)$ has a rational frequency ratio. We compared fourth order Runge–Kutta trajectories with exact solutions and adaptive integration. Phase shifts and radial displacements tested recovery in different directions. Reducing the integration step over a fixed interval measured numerical error. Appendix D gives the parameters and tolerances in model units.

We encoded a binary variable in the initial radial displacement and read it in two ways. One observation returns the exact displacement. The other returns the index of a bin of fixed width. For the phase pair, we compared two observations of the angles to determine when the observed value specifies its own future evolution (Appendix E).

3 Results

3.1 Tracking, Population Activity, and Spatial Entropy

The Figure 2c and 2d files share a tracking record within each numbered trial. Their position timestamps and both coordinate arrays match exactly. Establishing whether their spike clocks were synchronized would require acquisition metadata.

The archived population matrix pools both trials and an additional animal. The five files contain 23 cell recordings under 13 distinct labels. Merging spikes with repeated labels on one recording's time axis reproduces every entry in the matrix of 59,992 time bins by 13 labels. Its rows therefore combine activity from different recording times.

The stored persistence diagrams contain H_1 intervals for embeddings of the pooled matrix. The topology calculation stopped at H_1, so it supplies no corresponding H_2 intervals. Across the 13 spatial maps, the mean entropy of the normalized firing rates is 10.860 bits. Uniform rates across all 2,500 bins would give 11.288 bits. This comparison describes the distribution of firing rates across space (Appendix C).

3.2 Synchronization and Normal Attraction

The same toroidal phase coordinates support different orbit structures. With independent nonzero frequencies, a rational frequency ratio gives periodic orbits and an irrational ratio gives dense quasiperiodic orbits. Both phase directions remain neutral under perturbations. Under positive coupling, stable locking occurs when $|\Delta\omega| < 2\epsilon$, with

$$\delta_* = \arcsin\left(\frac{\Delta\omega}{2\epsilon}\right), \qquad \lambda_\delta = -\sqrt{4\epsilon^2 - (\Delta\omega)^2} < 0. \tag{4}$$

The stable locked set is a circle of fixed phase difference. It is a periodic orbit when the common frequency is nonzero and a circle of equilibria otherwise. The common phase remains neutral. At $|\Delta\omega| = 2\epsilon > 0$ the locking exponent vanishes. When $|\Delta\omega| > 2\epsilon$, the relative phase drifts. The complete branch and threshold analysis is given in Appendix A.

A radial displacement measures attraction to the invariant torus. For $M = \{r_1 = r_2 = 1\}$, its Euclidean distance satisfies

$$d(z(t), M) = e^{-\lambda t} d(z(0), M). \tag{5}$$

For identical initial phases on M, positive, zero, and negative radial rates give identical phase histories. A small transverse displacement decays, stays fixed, or grows, respectively, while the radii remain positive. With a positive rate, the torus attracts nearby states even when phase locking confines their limiting motion to a circle. If normal contraction dominates the tangent dynamics, the torus persists under sufficiently small autonomous perturbations [14]. Persistence of quasiperiodic motion on that torus requires additional hypotheses [15].

The simulations reproduce the predicted phase and amplitude changes. At $(\omega_1, \omega_2) = (1, \sqrt{2})$ and $\epsilon = 0.6$, the stable phase difference is 0.3524 rad and its exponent is -1.1262 per model time unit, to four decimal places. At $\epsilon = 0.1$, the mean slip rate is 0.3627 rad per model time unit. A phase shift persists under independent rotation. Under locking, the phase difference recovers while a common offset persists. The radial displacements decay at the predicted rate. Reducing the integration step gives approximately fourth order convergence until numerical precision limits the comparison (Figure 2).

3.3 Information Persistence and Coarse Dynamics

A readout sets the duration for which a contracting state reveals an encoded bit. Two distinct radial displacements remain distinguishable at every finite time under exact observation. A readout with fixed bin width eventually assigns both to the same bin. The differential entropy of a density in the normal coordinates also decreases under contraction. Estimating thermodynamic entropy flux would require physical measurements.

An observation supports autonomous coarse dynamics only if its evolution is consistent across states with the same observation. For fine and coarse flows F_t, G_t, a continuous

surjection π, and a time conversion $c > 0$, the required relation is

$$\pi \circ F_t = G_{ct} \circ \pi. \tag{6}$$

For nonzero coupling, the velocity of ϕ_1 depends on the unobserved phase. Its current value alone does not determine its future. The circular sum $(\phi_1 + \phi_2) \pmod{2\pi}$ does determine its own future, since it rotates at the constant rate $\omega_1 + \omega_2$. Thus two observations of the same system can differ in whether they satisfy Eq. (6).

4 Discussion

4.1 From Population Geometry to Neural Dynamics

Testing a neural attractor requires information about activity away from the proposed manifold. Yoon et al. used simultaneous recordings to study conserved relationships within grid cell populations [4]. The matrix reconstructed here pools separate recordings. Even a simultaneous record restricted to a torus could leave stability unresolved. In the oscillator model, every trajectory on the torus is consistent with attraction, neutrality, or repulsion.

A recovery experiment could estimate the manifold from simultaneous recordings of an identified grid cell module. The experiment would hold this estimate fixed and measure displacements in separate trials. Let $d(t)$ be the distance from the displaced population state to the estimated manifold in a specified activity metric. For $d(t_1), d(t_2) > 0$ and $t_2 > t_1$, the model predicts

$$\frac{\log[d(t_1)/d(t_2)]}{t_2 - t_1} = \lambda. \tag{7}$$

A positive rate indicates contraction, a zero rate neutrality, and a negative rate local repulsion. A shift along the torus can persist throughout the return. A neural test would compare decay rates with matched control trials and account for sensory and behavioral inputs. Distances near the measurement noise floor would require a separate estimate of uncertainty.

The motion that persists after recovery can occupy only part of the torus. In the locking regime, typical trajectories approach a circle within the attracting surface. Measuring recovery and tracking the phase difference would distinguish attraction to the surface from locking within it.

4.2 Entropy, Observation, and Continuity

The lifetime of an observed distinction depends on the readout. Under smooth contraction the exact states remain distinct, while a measurement with finite resolution can assign them the same value. An entropy measurement also requires a choice of variable and units. Spatial Shannon entropy describes the distribution of firing rates among bins. Differential entropy describes a density in specified coordinates. A thermodynamic estimate of neural cost would require a physical boundary and measurements of energy or matter exchanged across it (Appendix E.4).

Repeating a deterministic operation on a single observation preserves any equality already present in that observation. A subsequent measurement can supply a distinction that the first one missed. Predictive coding and Minimum Description Length motivate models of compression [16–18]. Whether successive biological layers shorten a description depends on their inputs and their codes. Relating those layers dynamically also requires a map that satisfies Eq. (6).

A theory of subjective continuity must specify which differences in neural state remain available to a later report or action. A human experiment could define a perceptual distinction in advance, specify a neural measurement, and test whether that measurement predicts the later report over a stated interval. The available rat recordings measure spatial firing and contain no reports of experience. Their relevance to this question is the distinction between persistence in a state and persistence in an observable.

5 Conclusion

Models with identical activity on a torus can respond differently to a displacement. Measuring the return rate distinguishes attraction, neutrality, and repulsion. The pooled grid cell matrix cannot provide that measurement. It requires a population recorded together through a displacement and its recovery.

In the contracting model, an exact observation preserves one bit at every finite time. A readout of fixed resolution eventually loses it. Testing continuity of a neural representation therefore requires tracking the displaced state and measuring which distinction a later observer can still recover.

A Geometry and Phase Dynamics

A.1 Periodic Spatial Coordinates

In an idealized grid code, two positions are equivalent if they differ by a lattice translation. Let $\Lambda = \mathbb{Z}\mathbf{e}_1 + \mathbb{Z}\mathbf{e}_2$ be a rank two lattice. The quotient $\mathbb{R}^2/\Lambda$ is a torus. Writing $\mathbf{x} = u_1\mathbf{e}_1 + u_2\mathbf{e}_2$ gives circular coordinates $\phi_i = 2\pi u_i \pmod{2\pi}$ (Figure 1). The periodic variables are the coefficients in the lattice basis.

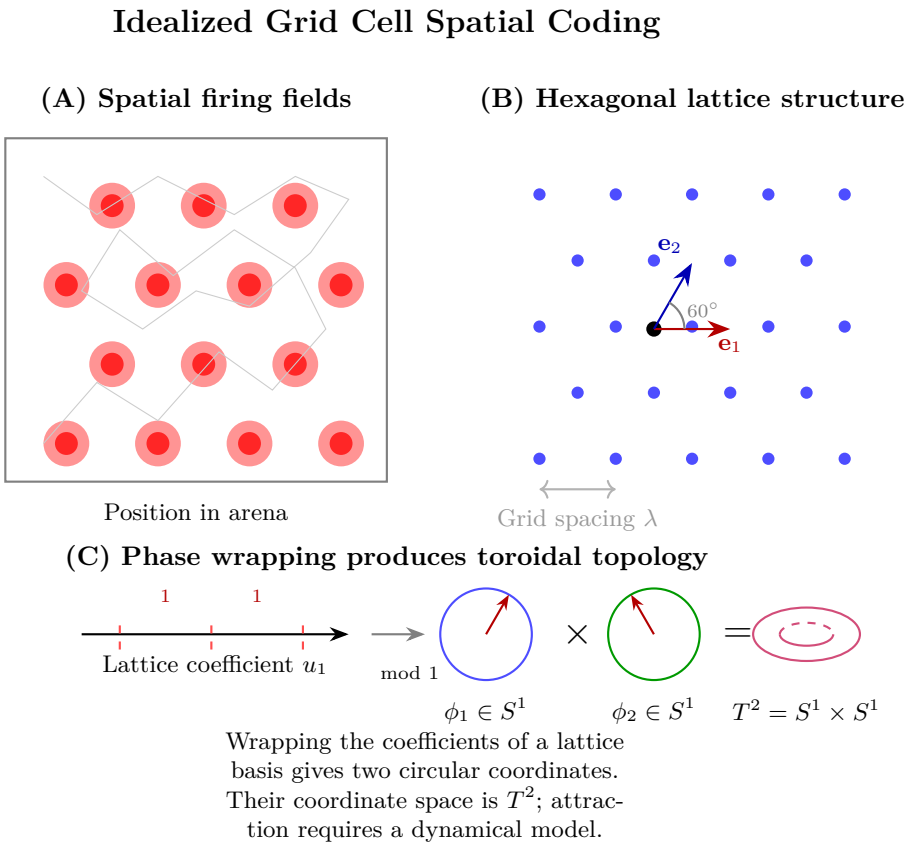


Figure 1: Periodic spatial coordinates. (A) Schematic grid cell firing fields. (B) Two lattice basis vectors separated by 60°. (C) Taking each lattice coefficient modulo one identifies positions that differ by a lattice translation and gives the quotient T^2.

To recover this quotient from firing rates, the population map must distinguish lattice phases and the recordings must sample them adequately. A single firing rate can have the same value at several phases. Spatial periodicity supplies candidate coordinates. Stability requires a description of the corresponding population dynamics.

Different spaces can have the same Betti numbers. Over a field, both T^2 and the wedge $S^1 \vee S^1 \vee S^2$ have $(\beta_0, \beta_1, \beta_2) = (1, 2, 1)$. Identifying a torus from these ranks requires additional assumptions or invariants. An unresolved H_2 feature leaves the topology open.

A.2 Circular Coordinates

A phase coordinate records position within an oscillation cycle [19, 20]. Consider a single idealized oscillator with phase variable $\theta_i(t)$ evolving as

$$\theta_i(t) = \omega_i t + \theta_{i,0}, \tag{A.1}$$

where $\theta_i(t)$ is the instantaneous phase of the i-th oscillator, ω_i is its angular frequency, and $\theta_{i,0}$ is the initial phase. Identifying phase modulo 2π places $e^{i\theta_i(t)}$ on S^1. For $\omega_i \neq 0$ the phase traverses the circle periodically. For $\omega_i = 0$ it stays fixed.

Two independently specifiable phases θ_1, θ_2 define the product space

$$T^2 = S^1 \times S^1, \tag{A.2}$$

of possible phase pairs. A given trajectory may visit only part of it. More generally, n independently specifiable circular coordinates define $T^n = (S^1)^n$ [20]. Constraints or locking can restrict the accessible or limiting set.

A standard embedding of T^2 in three dimensional Euclidean space has coordinates

$$x(\theta, \phi) = (R + r\cos\theta)\cos\phi, \tag{A.3}$$

$$y(\theta, \phi) = (R + r\cos\theta)\sin\phi, \tag{A.4}$$

$$z(\theta, \phi) = r\sin\theta, \tag{A.5}$$

For $R > r > 0$, R is the distance from the center of the torus to the center of the tube and r is the tube radius. The two angles $\theta, \phi \in [0, 2\pi)$ specify positions on this model surface. Inferring an embedding for neural activity would require population measurements.

A.3 Coupled Oscillator Dynamics on the Torus

Let $\phi_1, \phi_2 \in \mathbb{R}/2\pi\mathbb{Z}$. For C^1 coupling functions periodic in each phase, the equations

$$\dot{\phi}_1 = \omega_1 + \epsilon f_1(\phi_1, \phi_2), \tag{A.6}$$

$$\dot{\phi}_2 = \omega_2 + \epsilon f_2(\phi_1, \phi_2) \tag{A.7}$$

define a flow on T^2. Testing transverse attraction requires additional equations for states outside T^2.

At $\epsilon = 0$ with both frequencies nonzero, a rational ratio ω_1/ω_2 gives closed periodic trajectories. An irrational ratio gives trajectories dense in T^2. Perturbations in either phase direction remain neutral. For n independent rotations, density requires rational independence of the frequency vector, so $k \cdot \omega \neq 0$ for every nonzero $k \in \mathbb{Z}^n$. Coupling changes the equations and can change the orbit structure.

A.4 Locking Branches and the Critical Threshold

Subtracting the equations in Eq. (1) gives Eq. (2). On a lift, the mean phase $\psi = (\phi_1 + \phi_2)/2$ satisfies $\dot{\psi} = (\omega_1 + \omega_2)/2$. Changing one angular representative by 2π changes this mean by π, so the mean requires a choice of lift.

For $\epsilon > 0$ and $|\Delta\omega| < 2\epsilon$, $\dot{\delta} = 0$ gives the branch δ_* in Eq. (4) and the branch $\pi - \delta_*$ modulo 2π. Linearization at δ_* gives $\dot{\eta} = -2\epsilon \cos\delta_* \, \eta$. Since $\cos\delta_* > 0$, its exponent is $-\sqrt{4\epsilon^2 - (\Delta\omega)^2}$. The other branch has the opposite sign and is unstable.

The common phase remains neutral. A nonzero common frequency makes the stable circle a periodic orbit. At zero common frequency the circle consists of equilibria. At $|\Delta\omega| = 2\epsilon$ the locking exponent vanishes.

At the positive threshold $\Delta\omega = 2\epsilon > 0$, an exact solution with $\delta(0) = 0$ is

$$\delta(t) = 2\arctan\left(\frac{\epsilon t}{1 + \epsilon t}\right), \qquad \frac{\pi}{2} - \delta(t) \sim \frac{1}{\epsilon t}. \tag{A.8}$$

Setting $u = \tan(\delta/2)$ gives $\dot{u} = \epsilon(u-1)^2$ and yields Eq. (A.8). This solution approaches $\pi/2$ algebraically from below. A perturbation just above $\pi/2$ initially moves away and winds around the circle. The critical equilibrium therefore fails Lyapunov stability.

If $|\Delta\omega| > 2\epsilon$, the phase difference drifts. The mean unwrapped slip rate is

$$\langle\dot{\delta}\rangle = \operatorname{sgn}(\Delta\omega)\sqrt{(\Delta\omega)^2 - 4\epsilon^2}. \tag{A.9}$$

For positive detuning this follows by integrating one cycle, $T_\delta = \int_0^{2\pi}(\Delta\omega - 2\epsilon\sin u)^{-1}\,du = 2\pi/\sqrt{(\Delta\omega)^2 - 4\epsilon^2}$. The common and slip rotation rates can be commensurate even when the phases drift. In that case the trajectory closes.

The phase equations specify motion along the torus. The radial equations specify how nearby states move toward or away from it.

B Normal Attraction and Perturbation Calculations

A torus M attracts nearby states if it is invariant and trajectories from a neighborhood approach it in distance. Individual trajectories can visit only part of M. A sufficient condition for persistence under perturbations is that normal contraction dominate tangent growth in forward and backward time. Under the usual smoothness and compactness hypotheses, the normally hyperbolic manifold persists under sufficiently small autonomous C^1 perturbations of the vector field [14]. This condition places no nonresonance requirement on the frequencies along the torus.

Equation (3) extends the phase pair with positive radial coordinates r_1, r_2. In the embedding $(r_1\cos\phi_1, r_1\sin\phi_1, r_2\cos\phi_2, r_2\sin\phi_2) \in \mathbb{R}^4$, the set $M = \{r_1 = r_2 = 1\}$ is an invariant T^2 and

$$r_i(t) - 1 = (r_i(0) - 1)e^{-\lambda t}. \tag{B.1}$$

For positive radii, the nearest point on each unit circle has the same phase. Its squared distance is $(r_i - 1)^2$ in each coordinate plane. Summing gives $d(z, M)^2 = (r_1-1)^2 + (r_2-1)^2$ and Eq. (5). Changing λ changes the transverse response without altering the phase flow

on M. Positive λ gives normal attraction, zero gives neutral displacement, and negative λ gives local repulsion while the radii remain positive.

B.1 Normal Hyperbolicity and Internal Motion

Normal attraction can coexist with phase locking. For independent rotations the tangent exponents are zero. For the coupled pair, $\lambda > 2|\epsilon|$ is sufficient for normal contraction to dominate tangent growth. In the locking regime, typical trajectories approach the stable circle within the torus.

The phase Jacobian satisfies

$$J(\phi) = \epsilon \cos\delta \begin{pmatrix} -1 & 1 \\ 1 & -1 \end{pmatrix}, \qquad \|J(\phi)\|_2 \le 2|\epsilon|. \tag{B.2}$$

The forward and inverse tangent propagators are bounded by $e^{2|\epsilon|t}$ for $t \ge 0$. Normal displacements decay as $e^{-\lambda t}$. For $\lambda > 2|\epsilon|$, normal contraction dominates both tangent bounds. For every $\lambda > 0$, the explicit radial model attracts nearby states.

B.2 A Solvable Perturbation

Periodic radial forcing moves the invariant torus. With independent phases, replace Eq. (3) by

$$\dot{r}_i = -\lambda(r_i - 1) + a \sin\phi_i. \tag{B.3}$$

The attracting invariant torus is now the graph

$$r_i^*(\phi_i) = 1 + \frac{a(\lambda \sin\phi_i - \omega_i \cos\phi_i)}{\lambda^2 + \omega_i^2}. \tag{B.4}$$

For $|a| < \min_i \sqrt{\lambda^2 + \omega_i^2}$ the graph has positive radii. Substitution gives $\frac{d}{dt}(r_i - r_i^*(\phi_i)) = -\lambda(r_i - r_i^*(\phi_i))$. The radial displacement from the graph contracts exponentially. The distance to the graph is at most the norm of the displacement at the same phases, which proves attraction. The distance to the original unit torus can remain positive.

B.3 Scope of Persistence Results

Preserving an invariant surface and preserving quasiperiodic motion on that surface require different hypotheses. Perturbations can change rotation rates or introduce locking. Theorems that preserve quasiperiodic motion impose conditions on regularity, frequencies, nondegeneracy, and parameters [15]. Classical KAM results apply to Hamiltonian systems under their respective hypotheses [21–23]. Their application to a dissipative neural model would require a separate theorem.

C Data Processing and Provenance

C.1 Recording Inventory and Processing

The five MAT files contain position timestamps, coordinates, and spike timestamps from the Hafting recordings [1, 11]. We compared the three raw tracking arrays before filtering missing positions. SHA-256 digests match between the Figure 2c and 2d files within Trial 1 and within Trial 2. They differ between trials. The file hashes, cell labels, and matrix dimensions are in `outputs/claim_audit/empirical_provenance.json`. Matching tracking arrays identify a shared position record. We analyzed spike populations within each file.

The activity builder bins one selected recording at a time in complete 20 ms intervals. It divides spike counts by the bin width to obtain rates. Each command takes a recording name and writes to a separate output directory. To establish the origin of the pooled matrix, we also reproduced the earlier merging procedure and compared the resulting entries with the archive.

Spatial rate maps use a 50×50 position grid. The map generation code estimates occupancy time from the sample count and median tracking interval. It smooths spike counts and occupancy separately with a Gaussian kernel of width two spatial bins. It divides the smoothed quantities where occupancy exceeds 0.01 s. It assigns spikes to the nearest retained position timestamp. Irregular sampling and long gaps can bias this assignment.

C.2 Topology and Entropy Checks

We counted finite H_1 intervals in the stored persistence diagrams and measured their lifetimes in each embedding. We checked which homology dimensions and perturbations the code computed. Assessing the reliability of these features would require calibration of the full preprocessing, embedding, and filtration procedure.

Spatial entropy provides a separate description of the retained rate maps. For finite nonnegative rates r_b with positive total, we use

$$p_b = \frac{r_b}{\sum_j r_j}, \qquad H = -\sum_b p_b \log_2 p_b, \tag{C.1}$$

with $0 \log_2 0 = 0$. Each bin's weight is proportional to its firing rate. A distribution of spike locations would also depend on occupancy.

The population entropy routine normalizes activity across cell labels within each bin that has positive total activity. The temporal routine partitions samples into contiguous windows and measures the allocation of activity among them. Silent or missing distributions are undefined. The summaries give the number of defined bins. Both routines measure shares of activity rather than a process entropy rate.

C.3 Results by Data Product

C.3.1 Recording Provenance

The five files contain 23 cell recordings and 13 distinct labels (Table 1). Some labels occur in both trials. Counting labels alone cannot identify the number of distinct biological cells.

Table 1: Position samples and recorded cells in the five files. Position counts include nonfinite coordinates. Cell labels can repeat across files.

File	Position samples	Cell recordings
`Hafting_Fig2c_Trial1.mat`	59,596	7
`Hafting_Fig2c_Trial2.mat`	59,723	7
`Hafting_Fig2d_Trial1.mat`	59,596	3
`Hafting_Fig2d_Trial2.mat`	59,723	3
`rat_10925.mat`	89,652	3

Placing all five files on one recording's time axis and merging spikes with matching cell labels reproduces every entry of the $(59{,}992, 13)$ archived matrix. Its embeddings and population statistics are summaries of that pooled matrix. Separate recording selections prevent this pooling.

Earlier map exports reused filenames for repeated cell labels. The 13 surviving spatial maps have no one-to-one correspondence with all 23 cell recordings. They describe firing rates by location. Testing attraction requires the time course of a displaced population.

C.3.2 Persistence Diagrams and Controls

The pooled matrix has 32 finite H_1 intervals under principal component analysis (PCA) and 84 under the UMAP embedding. The maximum lifetimes are 0.668947 and 4.001927, respectively. UMAP changes distances, so these lifetimes use different metrics. The topology code computes H_0 and H_1 only.

The annotated manuscript shows rate maps, a UMAP embedding, and persistence barcodes attributed to Gardner modules. A separate archive preserves the barcode for Rat R, Day 1, Module 2 as a standalone image. It displays H_1 and H_2 intervals. Reproducing those intervals requires the missing numerical arrays and generating notebook. Reported parameter sweeps give inconsistent numerical values across archival sources. Quantifying a noise floor for the plotted gaps requires a control with matched sampling and known topology.

The available controls compare spatial smoothing, subsets of neurons, and shuffled columns of binned activity in the pooled matrix. They do not implement timestamp jitter. Reordering a list of timestamps leaves a histogram of spike counts unchanged. A timing surrogate must change the event times or their relationships. Spatial smoothing, temporal smoothing, and PCA whitening each act on a different quantity.

C.3.3 Spatial Entropy

The 13 spatial maps have mean entropy 10.860 bits under Eq. (C.1). Uniform rates across all 2,500 bins give $\log_2(2500) \approx 11.288$ bits. Applying the stated normalization to these maps does not reproduce a separately reported value of about 1.49 bits. Binning, smoothing, and treatment of missing positions all affect this calculation.

The pooled matrix has mean activity-share entropy 0.260240 bits. Pooling across trials rules out interpreting this value as the entropy of a simultaneous population. Both this value and the spatial-map statistic summarize a distribution of rates. Measuring information persistence requires observations at different times (Appendix E).

C.4 Sampling and Topological Calibration

Temporal sampling must match the frequency being tested. A 100 ms bin gives a sampling rate of 10 Hz and a Nyquist frequency of 5 Hz. It cannot resolve the full 4–8 Hz theta band. Gardner et al. used 10 ms spike count bins with separate smoothing and subsampling choices [2]. Sampling rate, filtering, and the target observable must be specified together.

Persistence diagrams require calibration against known topology under matched noise and sampling. Coverage, temporal dependence, distance metric, and preprocessing can change the recovered features. A high sample density by itself gives no general prediction for the number of spurious H_1 intervals.

D Numerical Methods and Verification

D.1 Simulation Design

All simulation parameters are in model units. The script `scripts/oscillator_validation.py` compares couplings $\epsilon = 0, 0.1, 0.6$ at $(\omega_1, \omega_2) = (1, \sqrt{2})$. A control uses $(1, 3/2)$ at $\epsilon = 0$. Each phase trajectory starts at $(0.2, 0.9)$. The fourth order Runge–Kutta (RK4) integration uses a fixed step $h = 0.02$ over 200 time units.

The radial simulations use $\lambda = 2$, initial radii $(0.7, 1.3)$, and forcing $a = 0$ or 0.1 over 20 time units. The unforced simulation also runs under phase locking. A short control reverses the normal rate while keeping the phase equations fixed.

D.2 Analytic Benchmarks

The independent trajectories agree with the exact rotation solution. The rational control returns after 4π. Density for the irrational ratio follows from the exact frequencies. A finite trajectory samples only part of the orbit. A phase shift of 0.2 rad persists under independent rotation. With locking, the phase difference recovers and a common offset persists.

At $\epsilon = 0.6$, Eq. (4) predicts $\delta_* \approx 0.3524$ rad and $\lambda_\delta \approx -1.1262$ per model time unit. At $\epsilon = 0.1$, Eq. (A.9) predicts a mean slip rate of 0.3627 rad per model time unit. Complete cycle times and independent quadrature agree with this rate. The radial trajectories agree with the solutions of Eqs. (3) and (B.4) (Figure 2).

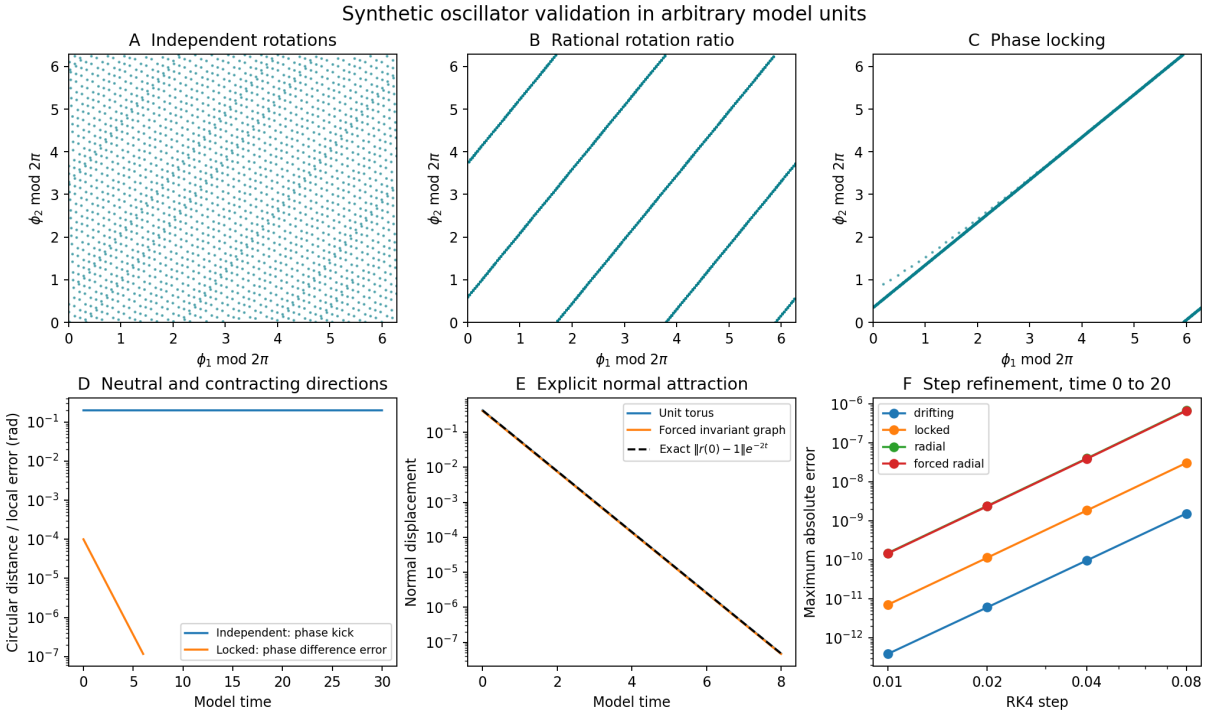


Figure 2: Simulated phase and radial trajectories in model units. The integration uses unwrapped angles. The plots wrap each angle into one period. Independent rotations and phase locking give different orbits in the same phase space. After a perturbation the common phase offset persists, while the stable phase difference and radial displacement recover. The step refinement panel compares RK4 with adaptive integration.

D.3 Numerical Error and Limits

We compared RK4 steps of $h = 0.08, 0.04, 0.02, 0.01$ over a fixed interval of 20 time units. The adaptive DOP853 reference used relative tolerance 10^{-12}, absolute tolerance 10^{-13}, and maximum step 0.1. A tighter DOP853 run and an independent RK45 integration checked the locked case.

The observed convergence order was approximately four before numerical precision limited further improvement. At $h = 0.01$, maximum absolute errors against the reference were approximately 3.9×10^{-13} for drift, 7.1×10^{-12} for locking, and 1.5×10^{-10} for both radial cases. Additional tests check exact solutions, stable and unstable branches, and neutral common phase. The repository includes the trajectories, convergence table, comparisons near the threshold, software versions, and a numerical summary.

The script `scripts/information_validation.py` compares the critical solution in Eq. (A.8) with adaptive integration. It also checks the observation rules in Appendix E.2 using the specified model parameters.

We evaluated the critical solution through $u/(1+u)$, with $u = \epsilon t$, to avoid cancellation near $t = 0$. The quantizer accepts finite bin labels below 2^{53}, a conservative bound for representing consecutive integers in double precision. Rounding near bin boundaries and underflow at very long times can affect the observed labels.

Short observation windows near the locking threshold make slow drift difficult to distinguish from slow recovery.

For the scalar normal mode, fixed step RK4 requires $|1 - z + z^2/2 - z^3/6 + z^4/24| < 1$, where $z = \lambda h$. The largest tested value, $z = 0.16$, satisfies this condition. The finite integrations cover deterministic trajectories over the stated intervals. Estimating effects of neural noise would require a noise model and data.

E Observation Maps and Entropy Calculations

E.1 Observation Classes and Recursive Processing

An observation map $\pi : X \to O$ assigns a readout to each system state. States are equivalent, $x \sim_\pi y$, when $\pi(x) = \pi(y)$. Equality makes this relation reflexive, symmetric, and transitive. The class at x is $[x]_\pi = \{y : \pi(y) = \pi(x)\}$. It contains several states when the readout assigns them the same value. A map of finite dimensional spaces can still be injective.

Repeated processing of the same observation gives the same output for equivalent states. For a deterministic map $R : O \to O$, $\pi(x) = \pi(y)$ implies $R^n(\pi(x)) = R^n(\pi(y))$ for every integer $n \geq 0$. A subsequent observation or an external input can distinguish states that the first measurement merged.

Two phase measurements define a toroidal readout. Given $\varphi_1, \varphi_2 : X \to S^1$, let $\Phi(x) = (\varphi_1(x), \varphi_2(x)) \in T^2$. Two states give the same pair exactly when both phases

agree. Whether this map covers the torus or embeds the state space depends on the choice of measurements. Attraction also requires equations of motion.

E.2 Contraction and Information Persistence

Normal contraction sets a rate for distance to a manifold. Distinct solutions of a smooth ordinary differential equation remain distinct at finite times on their common interval of existence. A finite or noisy measurement can still merge their readouts.

Encode an equiprobable binary label B in a radial displacement $x_B(0) \in \{0.25, 0.75\}$ and let $x_B(t) = e^{-2t}x_B(0)$. Exact observation distinguishes the labels at every finite time, so $I(B; x_B(t)) = 1$ bit.

A quantizer with bin width Δ reads $Q_\Delta(x) = \lfloor x/\Delta \rfloor$. Both displacements occupy bin zero for $t > \frac{1}{2}\log(0.75/\Delta)$. Their observed mutual information is then zero. At equality the larger displacement is exactly Δ and lies in bin one. Halving Δ delays entry into the shared zero bin by $(\log 2)/2$. For $\Delta = 0.1$ and 0.05, the respective thresholds are approximately 1.0075 and 1.3540 model time units.

The entropy of a continuous density decreases as contraction concentrates it. For a two dimensional normal displacement $X(t) = e^{-\lambda t}X(0)$ with a density and finite differential entropy, the change of variables formula gives

$$h(X(t)) = h(X(0)) + \log|\det(e^{-\lambda t}I_2)| = h(X(0)) - 2\lambda t, \tag{E.1}$$

in nats relative to fixed normal coordinates. The transformation remains invertible at every finite time. Calculating heat flux would require a physical model of energy exchange.

E.3 Consistency Across Scales

A fine flow F_t and a coarse flow G_t are compatible through a continuous surjection π if they satisfy Eq. (6) at a specified time conversion $c > 0$. This requires $\pi(x) = \pi(y)$ to imply $\pi(F_t(x)) = \pi(F_t(y))$. When π is differentiable, the respective vector fields f, g

must satisfy $D\pi_x f(x) = c\, g(\pi(x))$, by differentiation at $t = 0$. Continuity or smoothness of the induced flow on the observation space requires the corresponding topological or regularity assumptions.

Observing the first phase alone fails this condition when $\epsilon \neq 0$. Under $\pi(\phi_1, \phi_2) = \phi_1 \pmod{2\pi}$, the states $(0, 0)$ and $(0, \pi/2)$ give the same value. Their projected velocities are ω_1 and $\omega_1 + \epsilon$. At a short time $h > 0$, the lifted first phase of the second state exceeds that of the first by $\epsilon h + O(h^2)$. A later observation can therefore distinguish the states.

The circular sum $\sigma = (\phi_1 + \phi_2) \pmod{2\pi}$ satisfies $\dot{\sigma} = \omega_1 + \omega_2$ for every phase pair. Its initial value determines its entire trajectory. Applying this consistency test across molecular, neural, or behavioral scales would require readouts and measured dynamics at those scales.

E.4 Entropy and Physical Cost

Thermodynamic entropy requires a physical system boundary and specified exchanges. In units of k_B, its balance is [24]

$$\dot{S} = \dot{S}_i + \dot{S}_e, \qquad \dot{S}_i \geq 0. \tag{E.2}$$

At steady state, $\dot{S}_e = -\dot{S}_i \leq 0$. Net entropy export is positive in magnitude exactly when internal production is positive. Relating this balance to the topology of a neural state space would require a coupling between the physical and dynamical variables.

Restricting the set of represented states can shorten a code for those states. The change in code length depends on the code and its input distribution. Minimum Description Length and predictive coding motivate such conditional compression models [16–18]. Testing for compression through successive biological layers requires measurements at each layer. Comparing metabolic costs also requires independent measurements of physical exchanges.

F Quaternion Operators and Phase Coordinates

Three independently specifiable circular phases have coordinate space T^3. Unit quaternions instead form S^3, the double cover of $SO(3)$. No continuous embedding of T^3 into S^3 exists. Invariance of domain would make its image open, and compactness would make the image closed. Since S^3 is connected, such an embedding would be a homeomorphism. This contradicts $\pi_1(T^3) = \mathbb{Z}^3$ and $\pi_1(S^3) = 0$ [25].

A separately specified quaternion curve can describe rotation operators. For a pure quaternion angular velocity $\Omega(t)$,

$$\dot{q}(t) = \tfrac{1}{2}\Omega(t)q(t), \qquad |q(0)| = 1 \tag{F.1}$$

preserves $|q(t)| = 1$. To use these rotations on a phase model, one must specify an action on the phase states and an evolution law for $\Omega(t)$.

Ordinary phase translations commute. With $T_a(\phi) = \phi + a \pmod{2\pi}$, one has $T_a T_b = T_b T_a$. A model with noncommuting neural operators must specify the operators and their composition. An experiment could compare $A(B(x))$ and $B(A(x))$ from matched states. Changing the order of sampling also changes measurement times, which can produce an apparent dependence on order.

Phase amplitude coupling assigns different roles to a selected phase and amplitude [26]. The weighted phase lag index measures pairwise synchronization through the imaginary cross-spectrum [27]. Inferring a quaternion composition law from neural data would require additional measurements that control for delays, common input, amplitudes, and omitted state variables.

G Physical Assumptions Across Frequency Scales

Electromagnetic descriptions require assumptions about sources and material response. In a vacuum region without charges or currents, Maxwell's equations imply [28]

$$\nabla^2\mathbf{E} - \mu_0\epsilon_0\partial_t^2\mathbf{E} = 0. \tag{G.1}$$

For a homogeneous, isotropic, nondispersive Ohmic medium with constant μ, ϵ, σ, no free charge density, and no impressed current, the equation becomes

$$\nabla^2\mathbf{E} - \mu\epsilon\partial_t^2\mathbf{E} - \mu\sigma\partial_t\mathbf{E} = 0. \tag{G.2}$$

Applying these equations to neural tissue requires source currents, boundary conditions, and the tissue's dispersive response. A mechanism for toroidal neural dynamics would also require a measured interaction with the relevant state variables.

Compared with 10^3 Hz, frequencies of 10^9–10^{13} Hz are six to ten orders of magnitude higher. A channel gating time or inverse relaxation time can characterize an aperiodic process. A model linking frequency scales would need measured variables and specified interactions.

The band 0.1–10 THz corresponds to photon energies of 0.414–41.36 meV through $E = h\nu$. At 310 K, $k_BT \approx 26.7$ meV [29]. This thermal energy lies within the stated photon energy range. Molecular spectroscopy and Fröhlich's coherence proposal motivate studying collective modes [30, 31]. Establishing coherence, bond dissociation, or coupling to grid cell activity would require measurements specific to each process.

H Code and Data

The derivations use analytic solutions, with numerical checks for specified parameters. The archived Lean prototypes contain proof placeholders and undefined identifiers. They lack a pinned build and provide no formal verification of the claims.

Run `scripts/oscillator_validation.py` and `scripts/information_validation.py` to reproduce the oscillator trajectories and observation calculations. The script `scripts/audit_empirical_provenance.py` records the data inventory and checks the pooled matrix. The repository gives dependency versions, recording selection instructions, and test commands at https://github.com/ezernackchristopher97-cloud/ENTPTCCONPAPER26.

References

[1] Torkel Hafting, Marianne Fyhn, Sturla Molden, May-Britt Moser, and Edvard I. Moser. Microstructure of a spatial map in the entorhinal cortex. *Nature*, 436(7052): 801–806, 2005. doi: 10.1038/nature03721.

[2] Richard J. Gardner, Erik Hermansen, Marius Pachitariu, Yoram Burak, Nils A. Baas, Benjamin A. Dunn, May-Britt Moser, and Edvard I. Moser. Toroidal topology of population activity in grid cells. *Nature*, 602(7895):123–128, 2022. doi: 10.1038/s41586-021-04268-7.

[3] Erik Hermansen, David A. Klindt, and Benjamin A. Dunn. Uncovering 2-D toroidal representations in grid cell ensemble activity during 1-D behavior. *Nature Communications*, 15:5429, 2024. doi: 10.1038/s41467-024-49703-1.

[4] KiJung Yoon, Michael A. Buice, Caswell Barry, Robin Hayman, Neil Burgess, and Ila R. Fiete. Specific evidence of low-dimensional continuous attractor dynamics in grid cells. *Nature Neuroscience*, 16(8):1077–1084, 2013. doi: 10.1038/nn.3450.

[5] Si Wu, K. Y. Michael Wong, C. C. Alan Fung, Yuanyuan Mi, and Wenhao Zhang. Continuous attractor neural networks: candidate of a canonical model for neural information representation. *F1000Research*, 5:156, 2016. doi: 10.12688/f1000research.7387.1.

[6] Giovanni di Sarra, Siddharth Jha, and Yasser Roudi. The role of oscillations in grid cells' toroidal topology. *PLOS Computational Biology*, 21(1):e1012776, 2025. doi: 10.1371/journal.pcbi.1012776.

[7] Giovanni di Sarra, Siddharth Jha, and Yasser Roudi. Correction: The role of oscillations in grid cells' toroidal topology. *PLOS Computational Biology*, 21(10):e1013623, 2025. doi: 10.1371/journal.pcbi.1013623.

[8] György Buzsáki. *Rhythms of the Brain*. Oxford University Press, New York, 2006.

[9] Ryan T. Canolty, Erik Edwards, Sarang S. Dalal, Maryam Soltani, Srikantan S. Nagarajan, Heidi E. Kirsch, Mitchel S. Berger, Nicholas M. Barbaro, and Robert T. Knight. High gamma power is phase-locked to theta oscillations in human neocortex. *Science*, 313(5793):1626–1628, 2006. doi: 10.1126/science.1128115.

[10] Adriano B. L. Tort, Robert W. Komorowski, Joseph R. Manns, Nancy J. Kopell, and Howard Eichenbaum. Theta-gamma coupling increases during the learning of item-context associations. *Proceedings of the National Academy of Sciences*, 106(49): 20942–20947, 2009. doi: 10.1073/pnas.0911331106.

[11] Torkel Hafting and Centre for the Biology of Memory. Grid cell data hafting et al 2005. Data set, NIRD Research Data Archive, 2017. URL https://doi.org/10.11582/2017.00017.

[12] Yoshiki Kuramoto. *Chemical Oscillations, Waves, and Turbulence*. Springer, Berlin, 1984.

[13] Steven H. Strogatz. From Kuramoto to Crawford: exploring the onset of synchronization in populations of coupled oscillators. *Physica D*, 143(1–4):1–20, 2000. doi: 10.1016/S0167-2789(00)00094-4.

[14] Pierre Berger and Abed Bounemoura. A geometrical proof of the persistence of normally hyperbolic submanifolds. *Dynamical Systems*, 28(4):567–581, 2013. doi: 10.1080/14689367.2013.835386.

[15] Hendrik W. Broer, George B. Huitema, and Mikhail B. Sevryuk. *Quasi-Periodic Motions in Families of Dynamical Systems*, volume 1645 of *Lecture Notes in Mathematics*. Springer, 1996. doi: 10.1007/978-3-540-49613-7.

[16] Rajesh P. N. Rao and Dana H. Ballard. Predictive coding in the visual cortex: a functional interpretation of some extra-classical receptive-field effects. *Nature Neuroscience*, 2(1):79–87, 1999. doi: 10.1038/4580.

[17] Ming Li and Paul Vitányi. *An Introduction to Kolmogorov Complexity and Its Applications*. Springer, 3rd edition, 2008.

[18] Benjamin Prada, Shion Matsumoto, Abdul Malik Zekri, and Ankur Mali. Bridging predictive coding and MDL: A two-part code framework for deep learning, 2025. URL https://arxiv.org/abs/2505.14635v2. Preprint, version 2.

[19] Arthur T. Winfree. *The Geometry of Biological Time*, volume 12 of *Interdisciplinary Applied Mathematics*. Springer, 2nd edition, 2001. doi: 10.1007/978-1-4757-3484-3.

[20] Steven H. Strogatz. *Nonlinear Dynamics and Chaos: With Applications to Physics, Biology, Chemistry, and Engineering*. Westview Press, 2nd edition, 2015.

[21] Andrey N. Kolmogorov. On conservation of conditionally periodic motions under small perturbations of the Hamiltonian. *Doklady Akademii Nauk SSSR*, 98:527–530, 1954.

[22] Vladimir I. Arnold. Proof of a theorem of A. N. Kolmogorov on the invariance of quasi-periodic motions under small perturbations of the Hamiltonian. *Russian Mathematical Surveys*, 18(5):9–36, 1963.

[23] Jürgen Moser. On invariant curves of area-preserving mappings of an annulus. *Nachrichten der Akademie der Wissenschaften in Göttingen, Mathematisch-Physikalische Klasse*, pages 1–20, 1962.

[24] Gregoire Nicolis and Ilya Prigogine. *Self-Organization in Nonequilibrium Systems: From Dissipative Structures to Order through Fluctuations*. Wiley-Interscience, New York, 1977.

[25] Allen Hatcher. *Algebraic Topology*. Cambridge University Press, 2002. URL https://pi.math.cornell.edu/~hatcher/AT/AT.pdf.

[26] Adriano B. L. Tort, Robert Komorowski, Howard Eichenbaum, and Nancy Kopell. Measuring phase-amplitude coupling between neuronal oscillations of different frequencies. *Journal of Neurophysiology*, 104(2):1195–1210, 2010. doi: 10.1152/jn.00106.2010.

[27] Martin Vinck, Robert Oostenveld, Marijn van Wingerden, Francesco Battaglia, and Cyriel M. A. Pennartz. An improved index of phase-synchronization for electrophysiological data in the presence of volume-conduction, noise and sample-size bias. *NeuroImage*, 55(4):1548–1565, 2011. doi: 10.1016/j.neuroimage.2011.01.055.

[28] David Tong. Lectures on electromagnetism. University of Cambridge lecture notes, 2015. URL https://www.damtp.cam.ac.uk/user/tong/em.html.

[29] National Institute of Standards and Technology. Meet the constants, 2018. URL https://www.nist.gov/si-redefinition/meet-constants. SI defining values for the Planck constant, elementary charge, and Boltzmann constant; accessed November 3, 2026.

[30] Andrea G. Markelz. Terahertz dielectric sensitivity to biomolecular structure and function. *IEEE Journal of Selected Topics in Quantum Electronics*, 14(1):180–190, 2008. doi: 10.1109/JSTQE.2007.913424.

[31] Herbert Fröhlich. Long-range coherence and energy storage in biological systems. *International Journal of Quantum Chemistry*, 2(5):641–649, 1968. doi: 10.1002/qua.560020505.